DOXYCYCLINE

Avoid <8 yr & pregnancy

Take upright with water

CHEMICAL NAME (4S,4aR,5S,5aR,6R,12aS)-4-(dimethylamino)-3,5,10,12,12a-pentahydroxy-6-methyl-1,11-dioxo-1,4,4a,5,5a,6,11,12a-octahydrotetracene-2-carboxamide (hyclate or monohydrate)

CLASS Tetracycline antibiotic (long-acting)

TRADE NAMES Vibramycin · Doxylin · Doryx · Vibra-Tabs

THERAPEUTIC USES / INDICATIONS

- Respiratory infections: CAP (atypical cover), COPD exacerbation.
- Brucellosis (with rifampicin/streptomycin), rickettsial infections, Q fever, Lyme disease.
- Chlamydia, PID, syphilis (penicillin allergy).
- Acne, rosacea; malaria prophylaxis; cholera; anthrax; leptospirosis.

ADMINISTRATION

- **PO** with a full glass of water, sitting or standing upright for 30 min; not at bedtime (oesophageal ulcers).
- May be taken with food (reduces nausea); avoid milk, antacids, iron, calcium within 2–3 h.
- **IV** infusion (where available) over 1–4 h, protected from light.

DOSAGES

- **Adult:** 200 mg on day 1, then 100 mg daily (severe: 100 mg every 12 h).
- **Chlamydia:** 100 mg twice daily for 7 days.
- **Brucellosis:** 100 mg twice daily for 6 weeks (combination).
- **Malaria prophylaxis:** 100 mg daily (start 1–2 days before, continue 4 weeks after).
- **Children ≥8 yr:** 2.2 mg/kg every 12 h (max 100 mg/dose).

PHARMACOKINETICS

- **Peak:** 2–3 h. Bioavailability ~95%.
- Lipid-soluble; good tissue penetration.
- Excreted mainly in faeces (safe in renal impairment). **t½:** 16–22 h.

PHARMACODYNAMICS

- Binds **30S ribosomal subunit** → blocks tRNA attachment → bacteriostatic.
- Broad spectrum incl. intracellular organisms (*Chlamydia*, *Rickettsia*, *Brucella*, *Mycoplasma*).
- Anti-inflammatory and anti-collagenase effects (acne, rosacea).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, epigastric pain.
- Photosensitivity.
- Vaginal candidiasis.

ADVERSE EFFECTS (SERIOUS)

- Oesophageal ulceration (tablets lodging).
- Tooth discolouration and enamel hypoplasia; impaired bone growth in children <8 yr and fetuses.
- Benign intracranial hypertension (headache, visual disturbance).
- Hepatotoxicity; *C. difficile*.

CONTRAINDICATIONS & CAUTIONS

- Tetracycline hypersensitivity.
- Pregnancy (after 15 weeks), breastfeeding, children <8 yr (except short courses for life-threatening infections e.g. rickettsial).
- **Caution:** myasthenia gravis, hepatic impairment, SLE.

KEY INTERACTIONS

- **Antacids, Ca, Mg, Fe, Zn, bismuth, dairy:** chelation → ↓ absorption.
- **Warfarin:** ↑ INR.
- **Isotretinoin, retinoids:** ↑ intracranial hypertension.
- **Carbamazepine, phenytoin, rifampicin:** ↓ doxycycline levels.

NURSING MANAGEMENT

- Check pregnancy status and age.
- Give with full glass of water; patient upright 30 min after.
- Separate from minerals and dairy.
- Monitor for headache with visual changes.
- Monitor LFTs and symptoms in long courses.

PATIENT EDUCATION

- Swallow with a full glass of water; stay upright 30 min; not just before bed.
- Avoid strong sunlight; use sunscreen.
- Do not take with milk, antacids or iron.
- Report severe headache or blurred vision.

KEY POINTS

- Tetracycline; bacteriostatic; 30S inhibitor.
- Not <8 yr or in pregnancy (teeth/bone).
- Brucellosis, rickettsia, chlamydia, atypicals.
- Upright with water; photosensitivity.

SOURCES CHECKED **[177]** Doxycycline 100mg Capsules (UK SmPC) **[178]** Doxycycline, oral (US label) **[179]** Doxycycline, IV (US label)

Full references p. 432–457